

Universality of Local Polytope

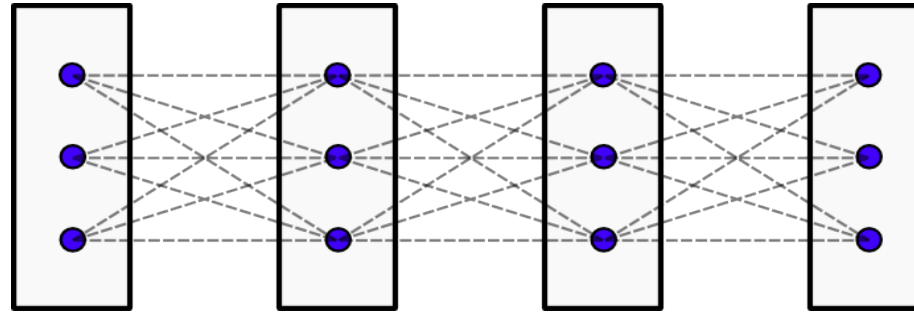
Daniel Pruša and Tomas Werner, 2014:

Theorem:

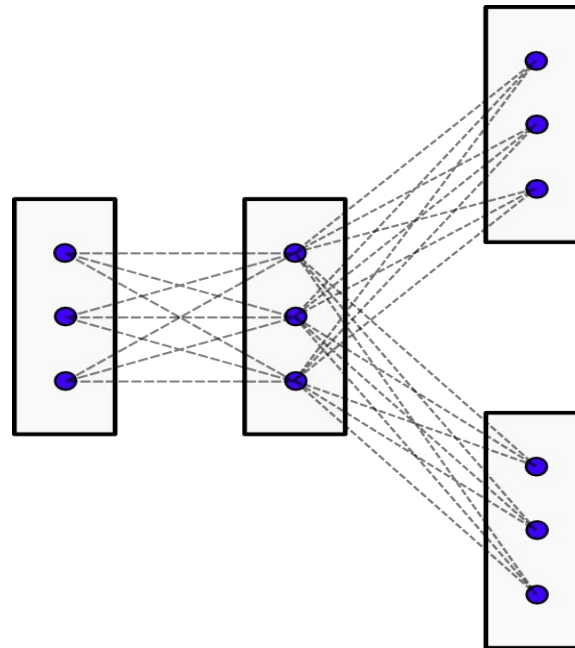
"Any linear program reduces to the local polytope relaxation of certain graphical model with 3 labels in linear time."

Acyclic Graphical Models

Chain-structured graphical model

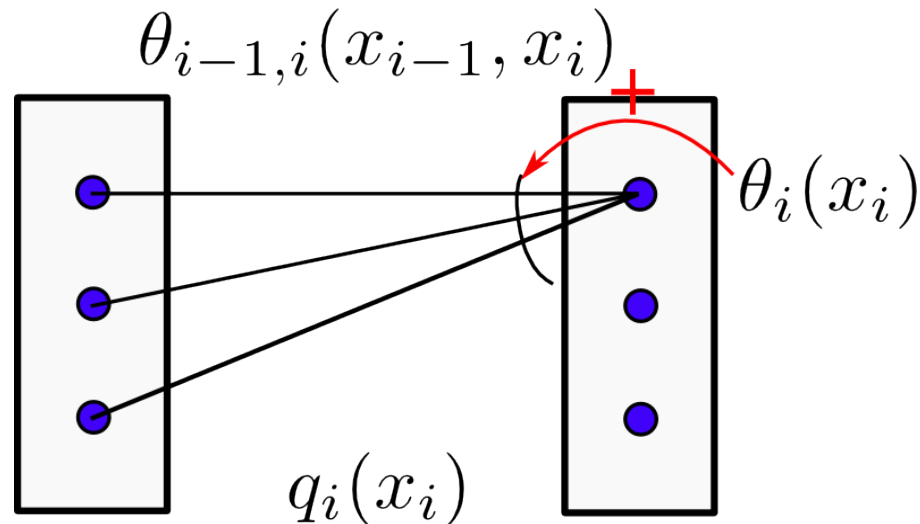


Tree-structured graphical model



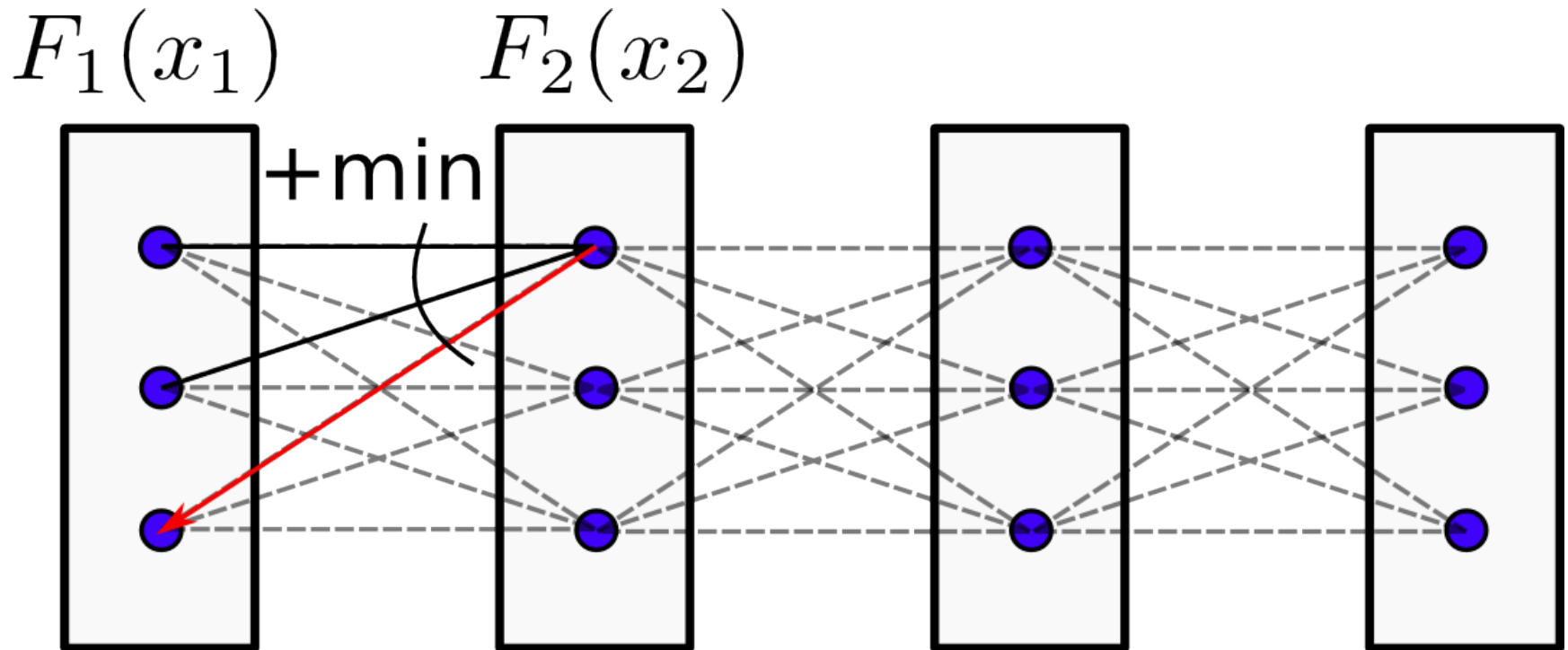
Chain-Structured Graphical Models

$$\begin{aligned} \min_{x_1, \dots, x_n} \underbrace{\theta_1(x_1)}_{q_1(x_1)} + \sum_{i=2}^n \underbrace{\left(\theta_{i-1,i}(x_{i-1}, x_i) + \theta_i(x_i) \right)}_{q_i(x_{i-1}, x_i)} \\ = \min_{x_1, \dots, x_n} q_1(x_1) + \sum_{i=2}^n q_i(x_{i-1}, x_i) \end{aligned}$$



Chain-Structured Graphical Models

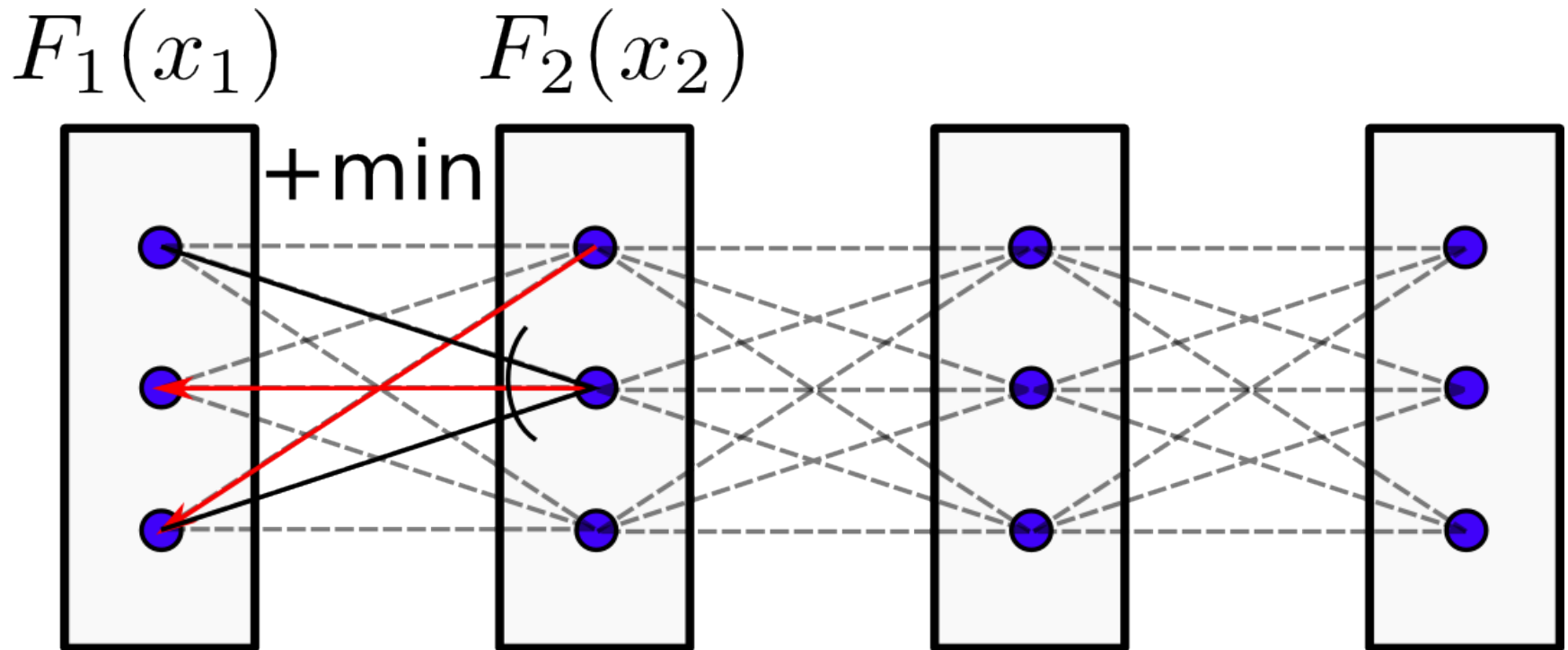
Compute the shortest path from the left:



$$F_i(s) := \min_{t \in i-1} F_{i-1}(t) + q_i(s, t)$$

Chain-Structured Graphical Models

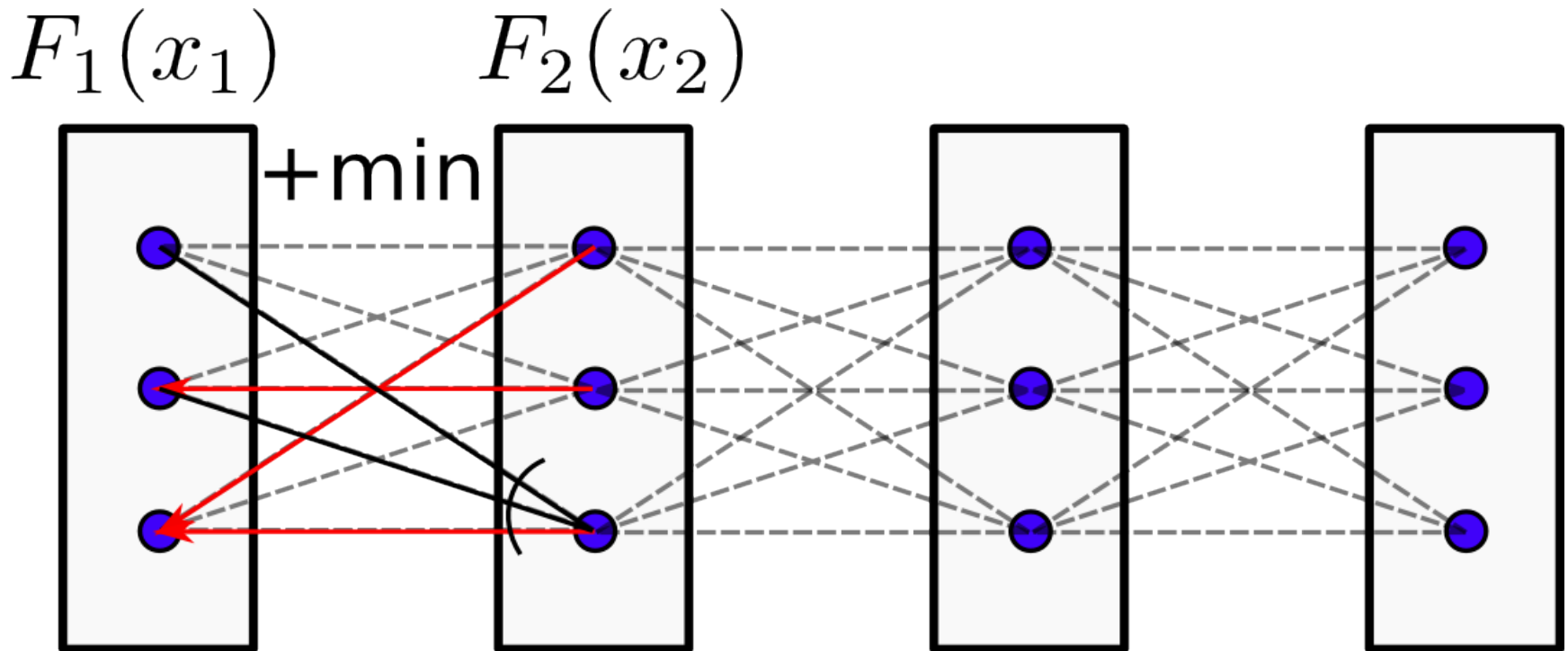
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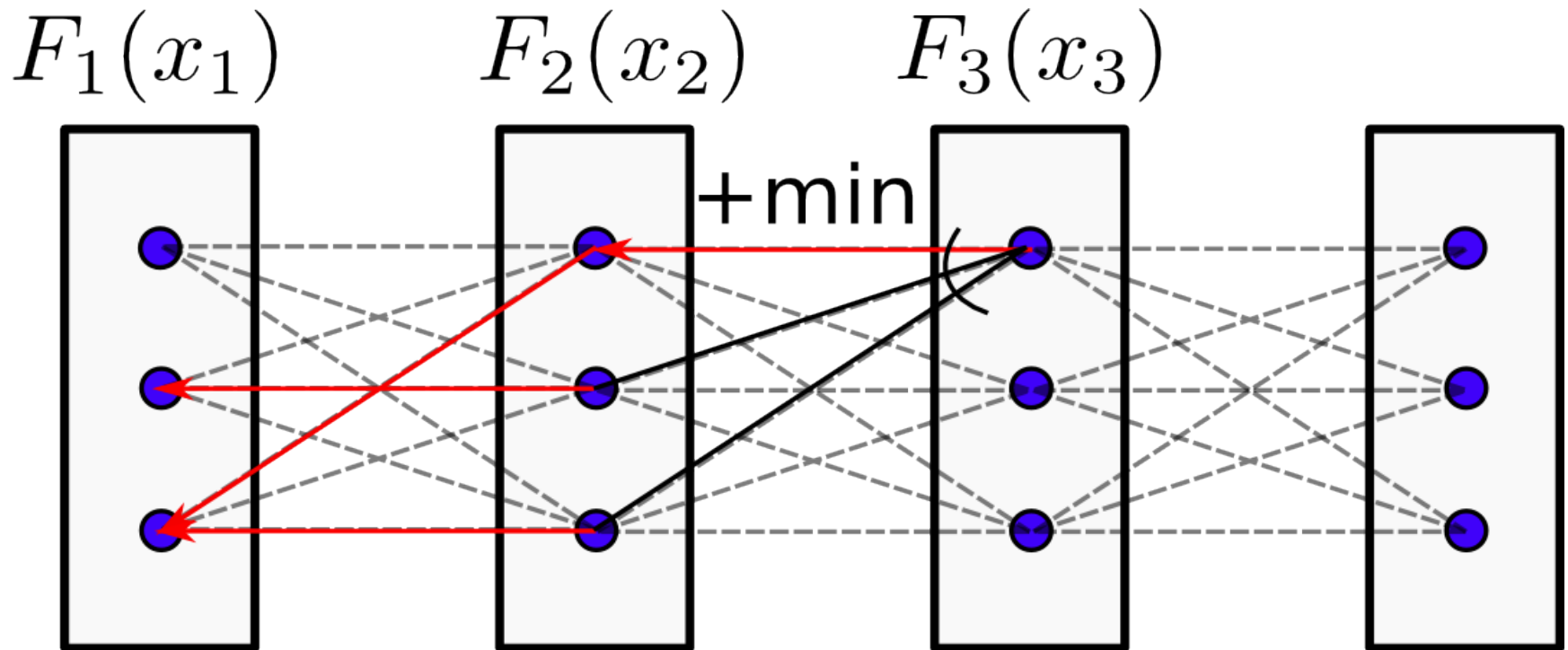
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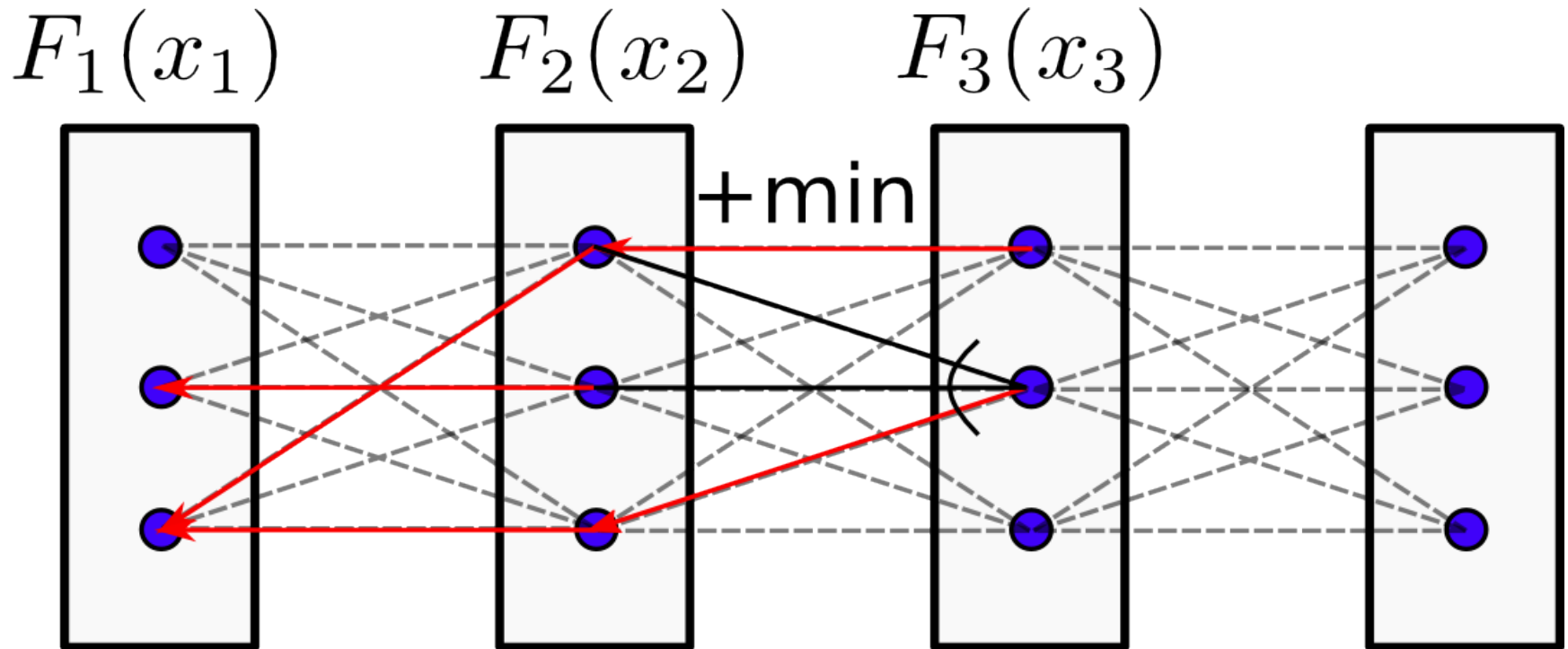
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Chain-Structured Graphical Models

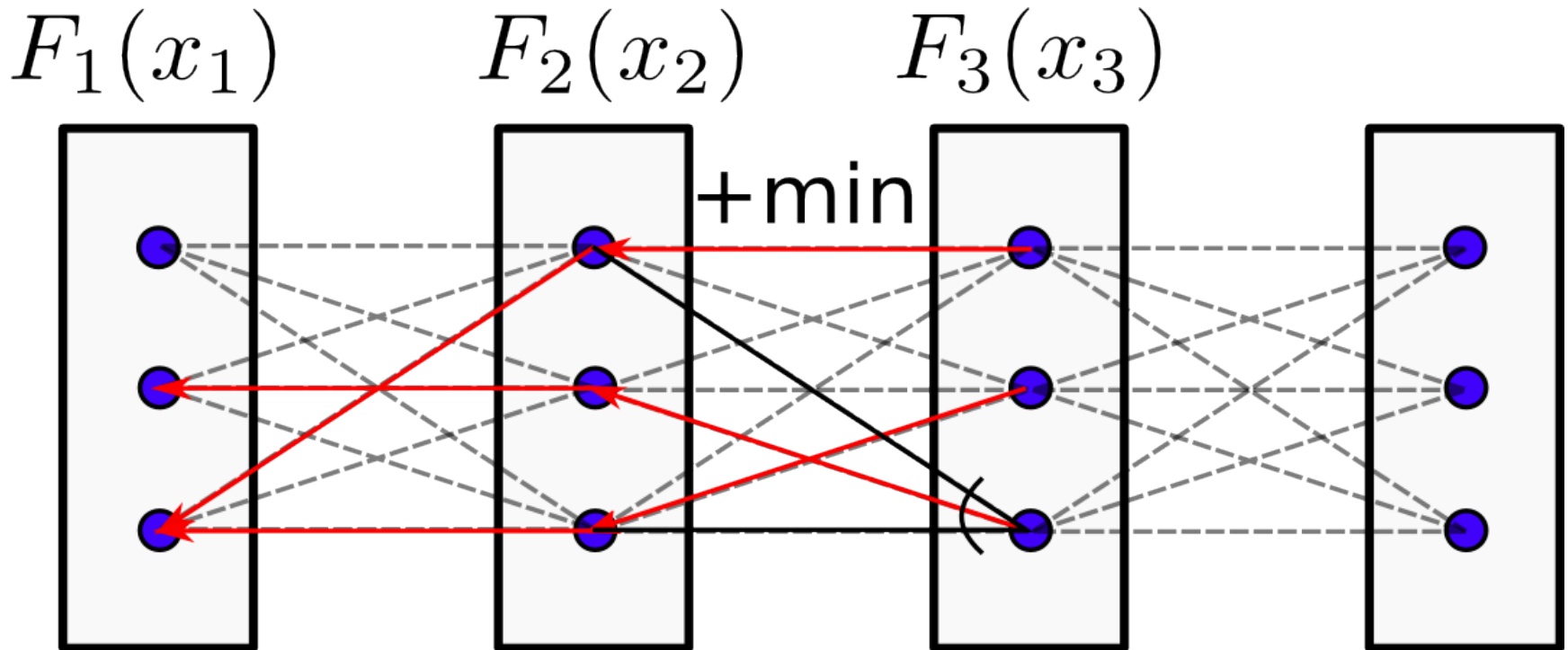
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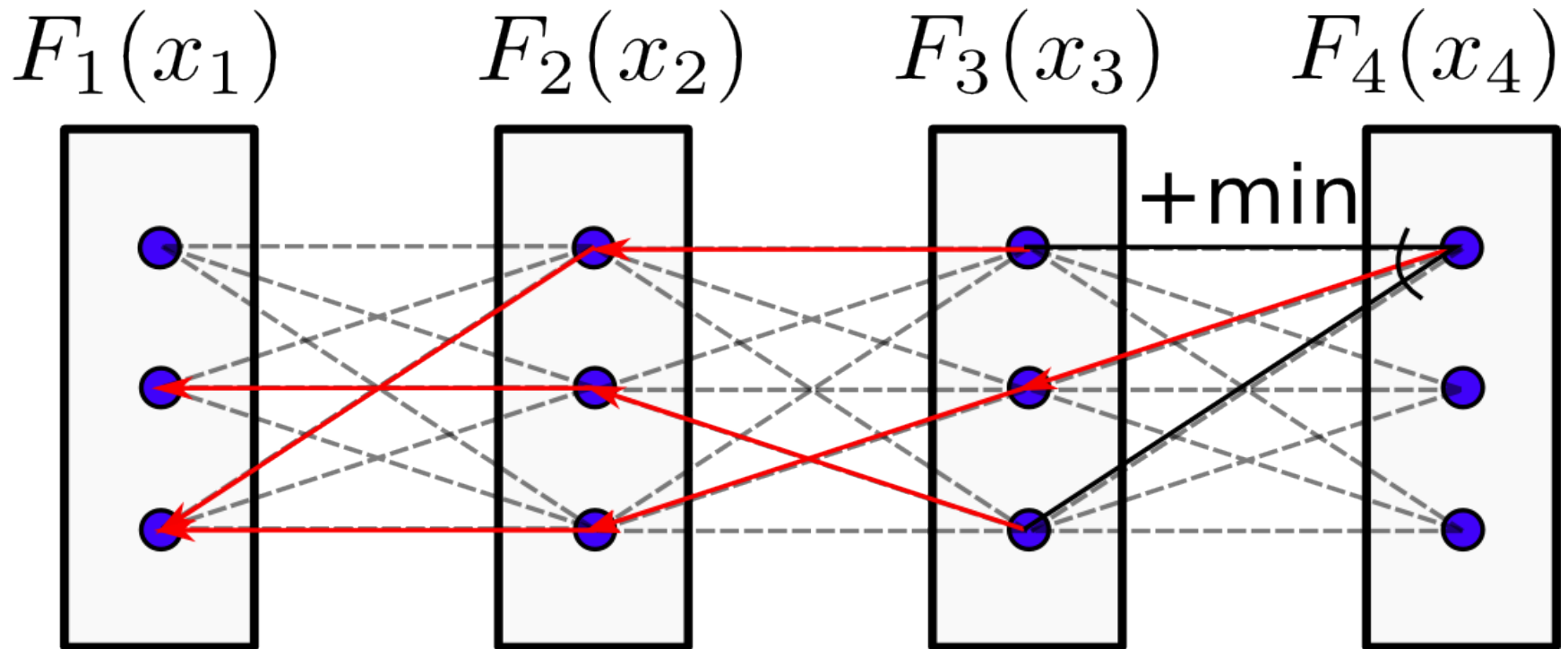
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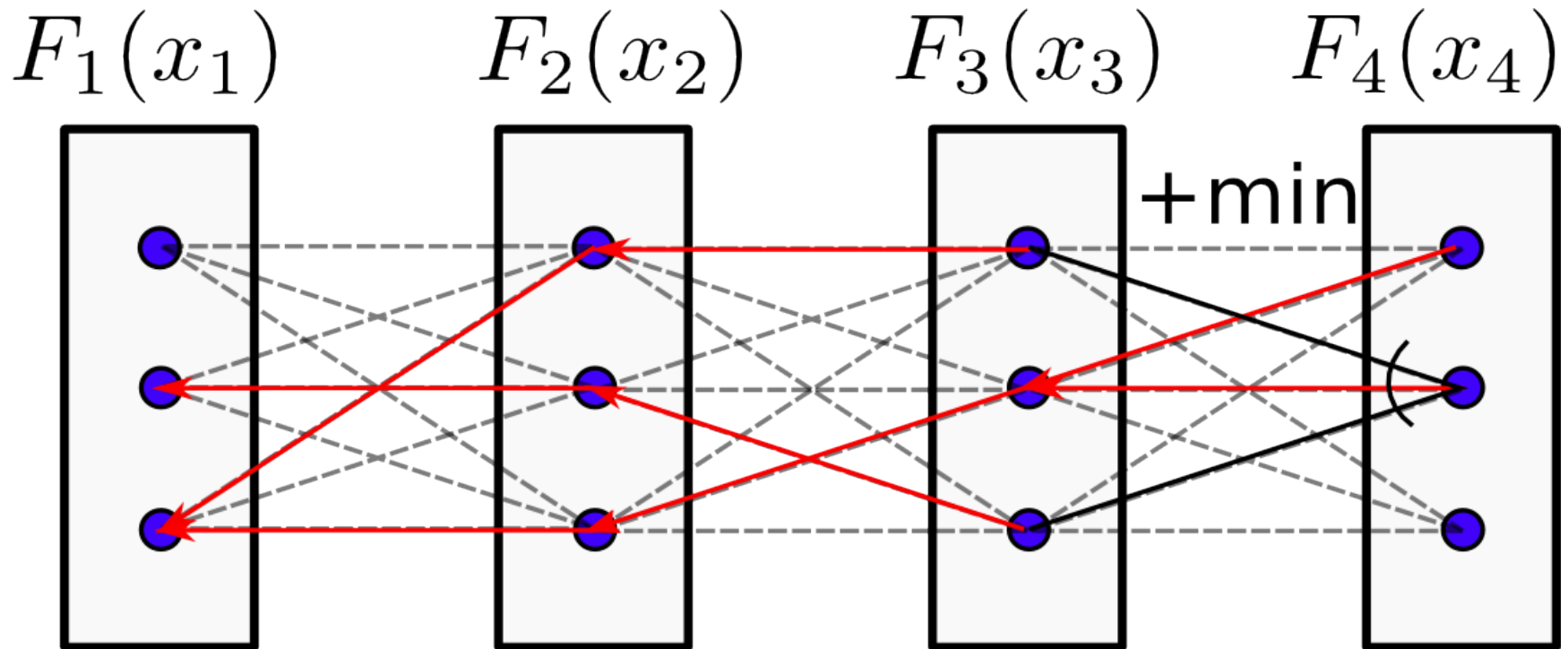
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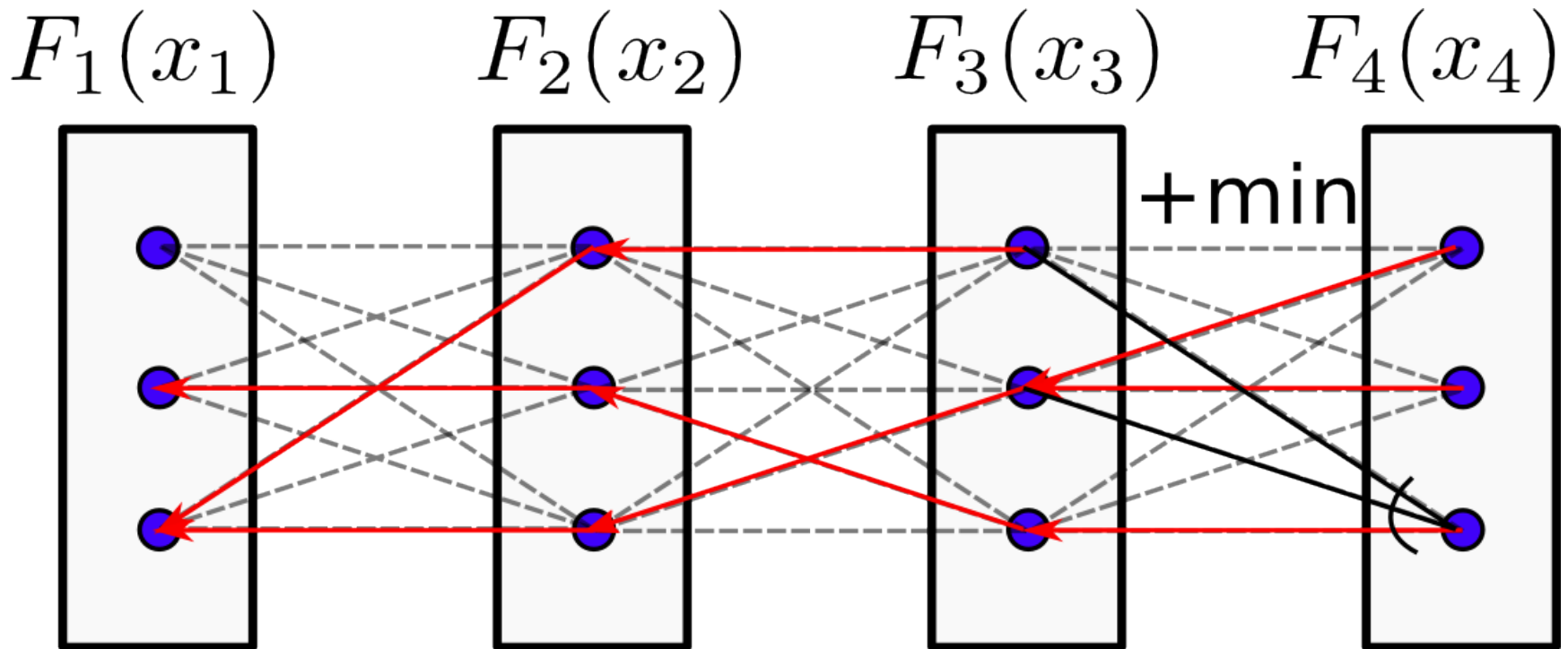
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Chain-Structured Graphical Models

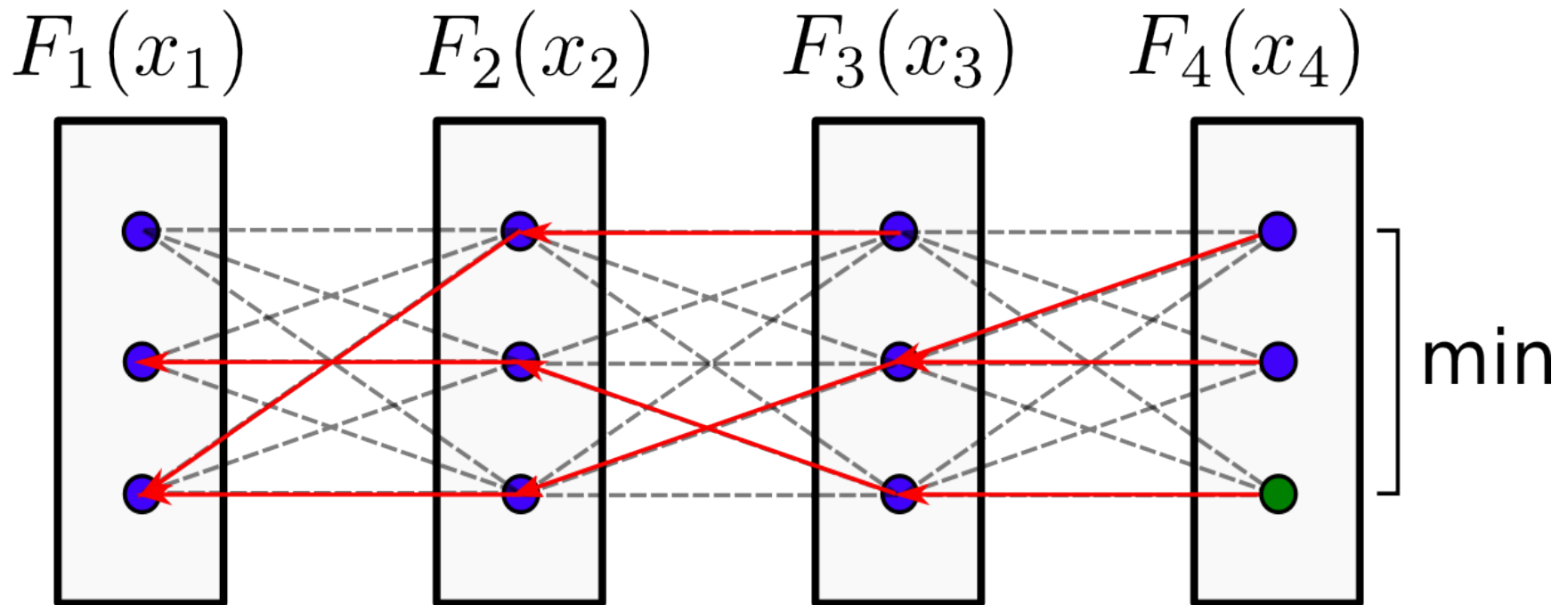
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Chain-Structured Graphical Models

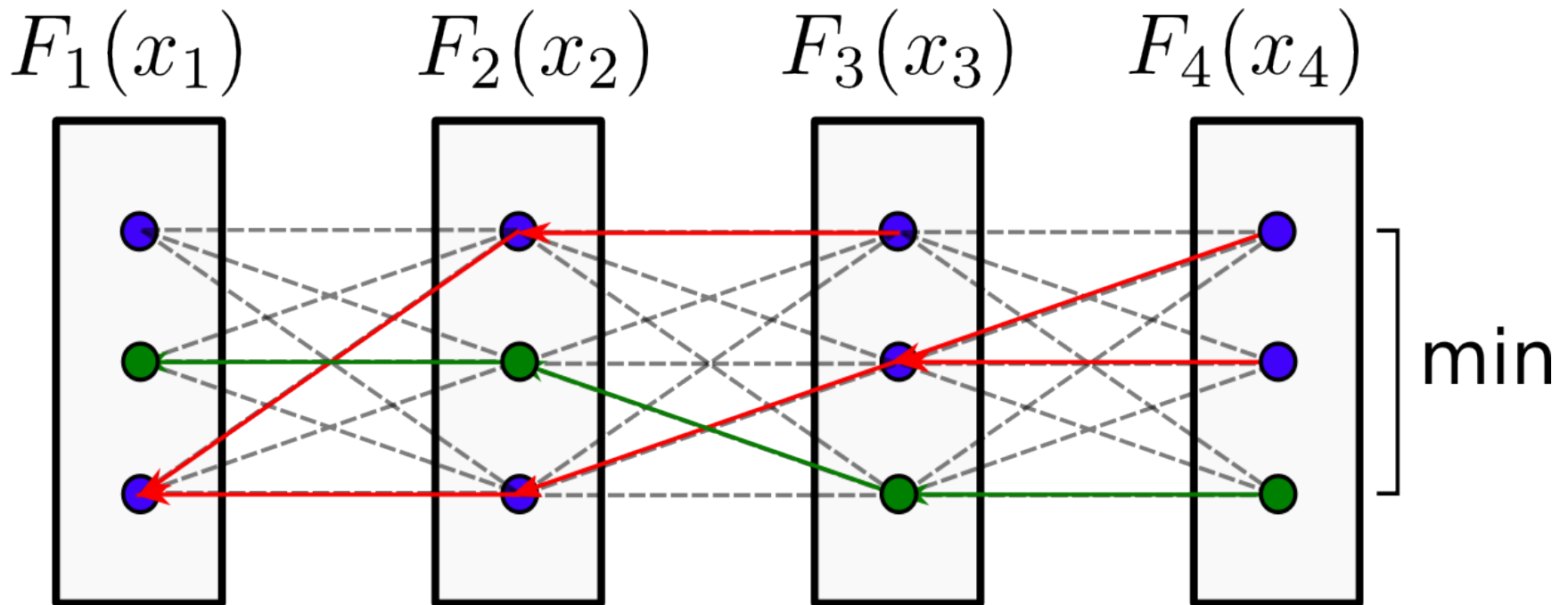
Select a label with a minimal weight:



$$x_n^* = \arg \min_{s \in X_n} F_n(s)$$

Chain-Structured Graphical Models

Track back the labeling:



$$x_n^* = \arg \min_{s \in X_n} F_n(s)$$

Chain-Structured Graphical Models

$$\begin{aligned} & \min_{x_1, \dots, x_n} q_1(x_1) + \sum_{i=2}^n q_i(x_{i-1}, x_i) \\ &= \min_{x_2, \dots, x_n} \left(\underbrace{\min_{x_1 \in_1} \left[\overbrace{q_1(x_1) + q_2(x_1, x_2)}^{F_1(x_1)} \right]}_{F_2(x_2)} + \sum_{i=3}^n q_i(x_{i-1}, x_i) \right) \\ &= \min_{x_2, \dots, x_n} \left(F_2(x_2) + \sum_{i=3}^n q_i(x_{i-1}, x_i) \right) \\ &= \min_{x_3, \dots, x_n} \left(\underbrace{\min_{x_2 \in_2} \left[F_2(x_2) + q_3(x_2, x_3) \right]}_{F_3(x_3)} + \sum_{i=4}^n q_i(x_{i-1}, x_i) \right) \\ &= \min_{x_3, \dots, x_n} \left(F_3(x_3) + \sum_{i=4}^n q_i(x_{i-1}, x_i) \right) = \dots \quad (1) \end{aligned}$$

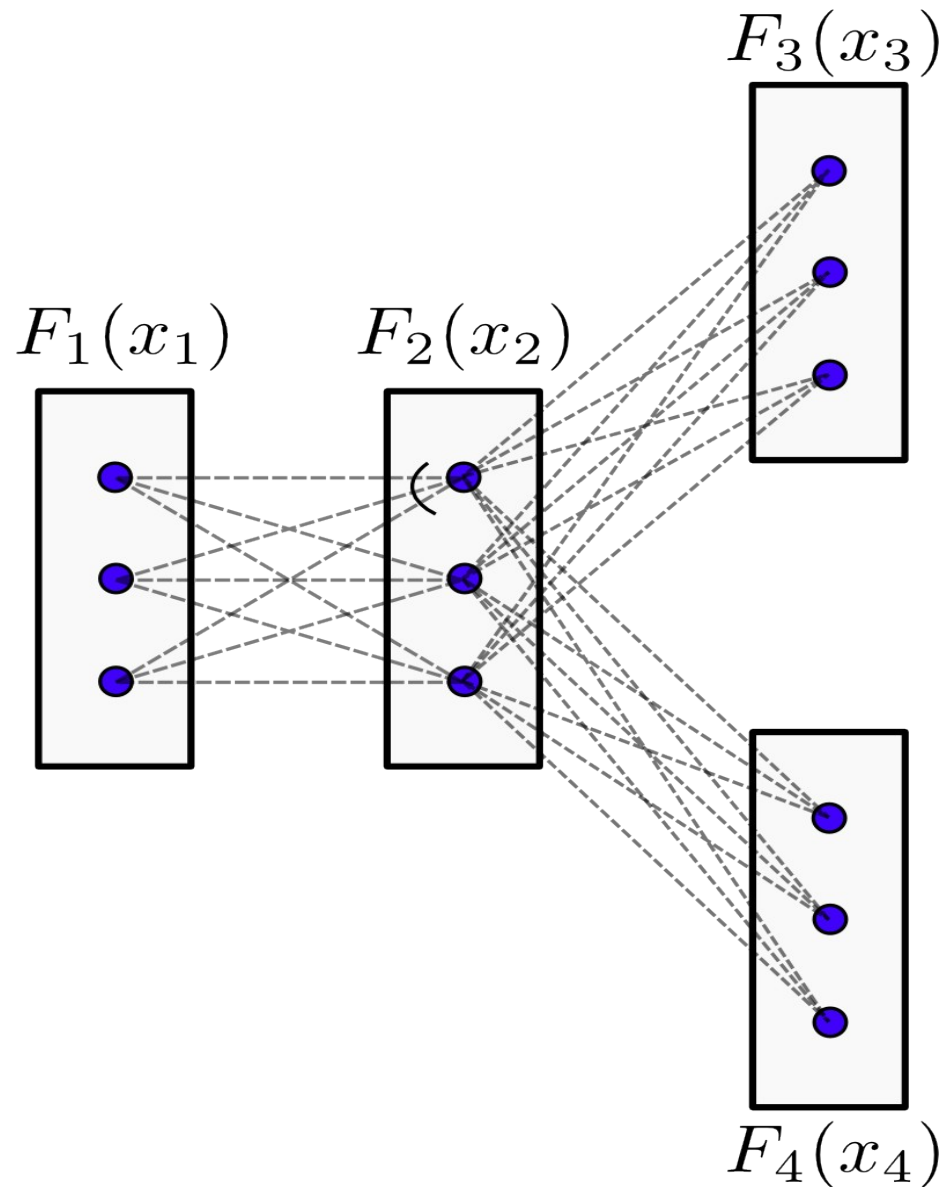
Algorithm complexity ?

Algorithm complexity ?

$|X_0|$ – number of labels

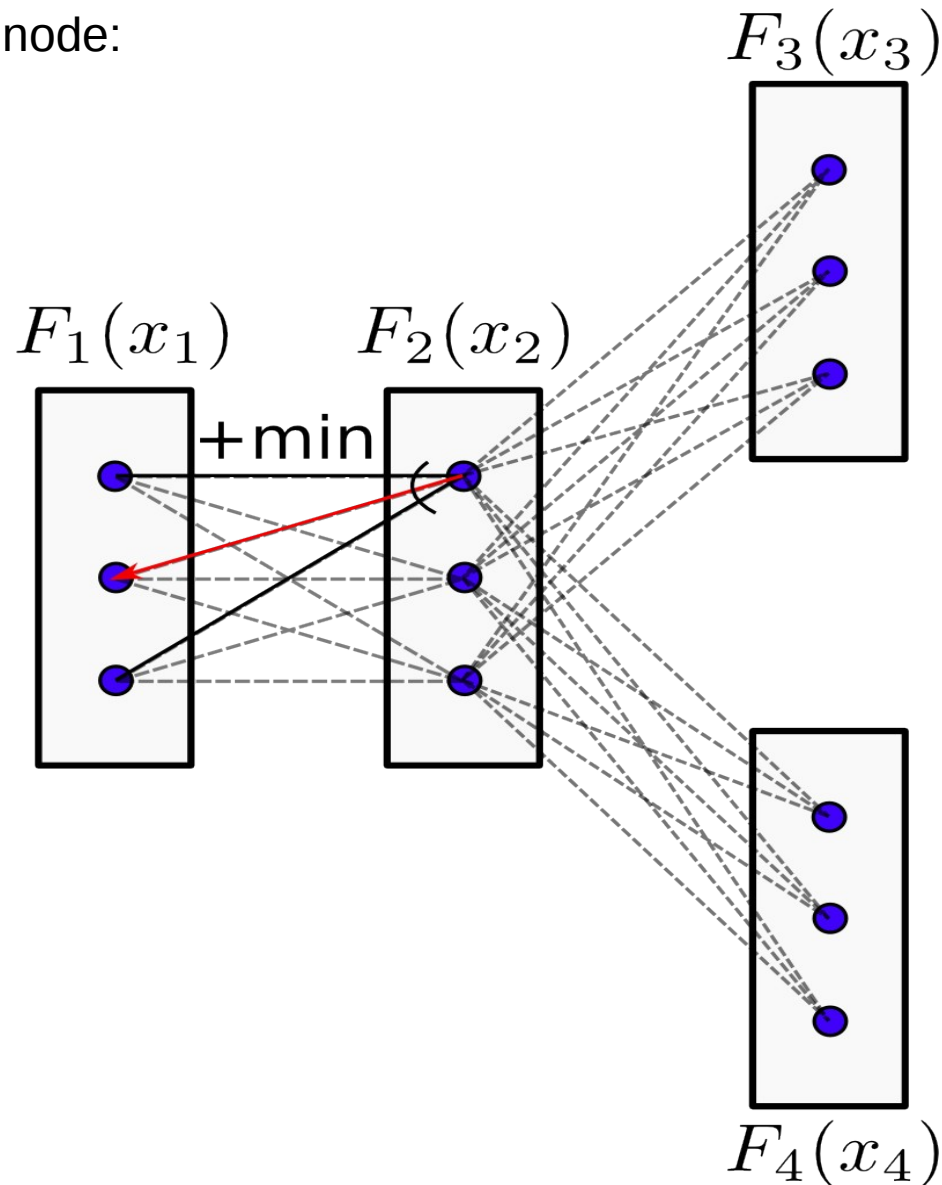
- 1) $O(|X_0| \cdot |V|)$
- 2) $O(|X_0|^2 |V|^2)$
- 3) $O(|X_0|^2 |V|)$
- 4) $O(|X_0| \cdot |V|^2)$
- 5) $O(\log |X_0| \cdot |V|)$
- 6) None is correct

Tree-Structured Graphical Models

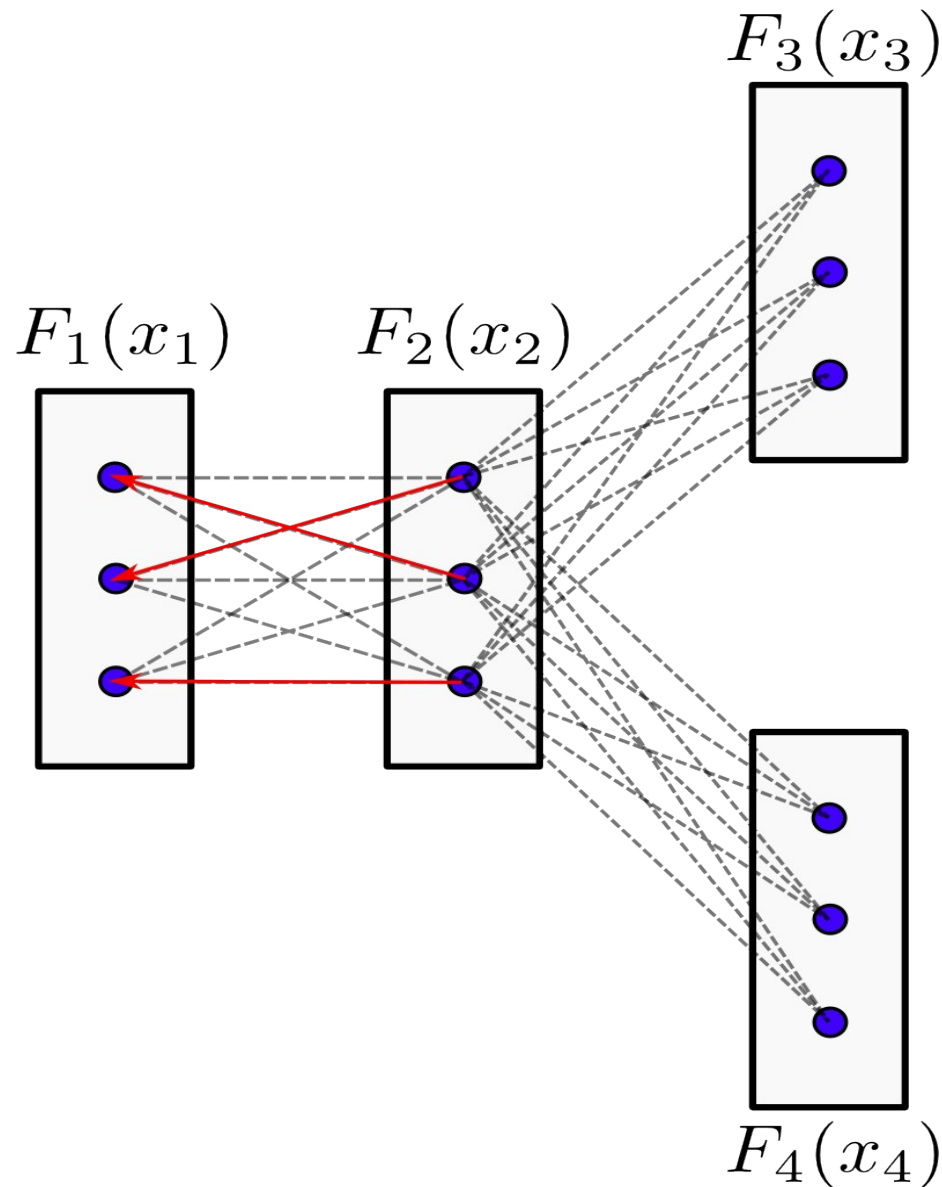


Tree-Structured Graphical Models

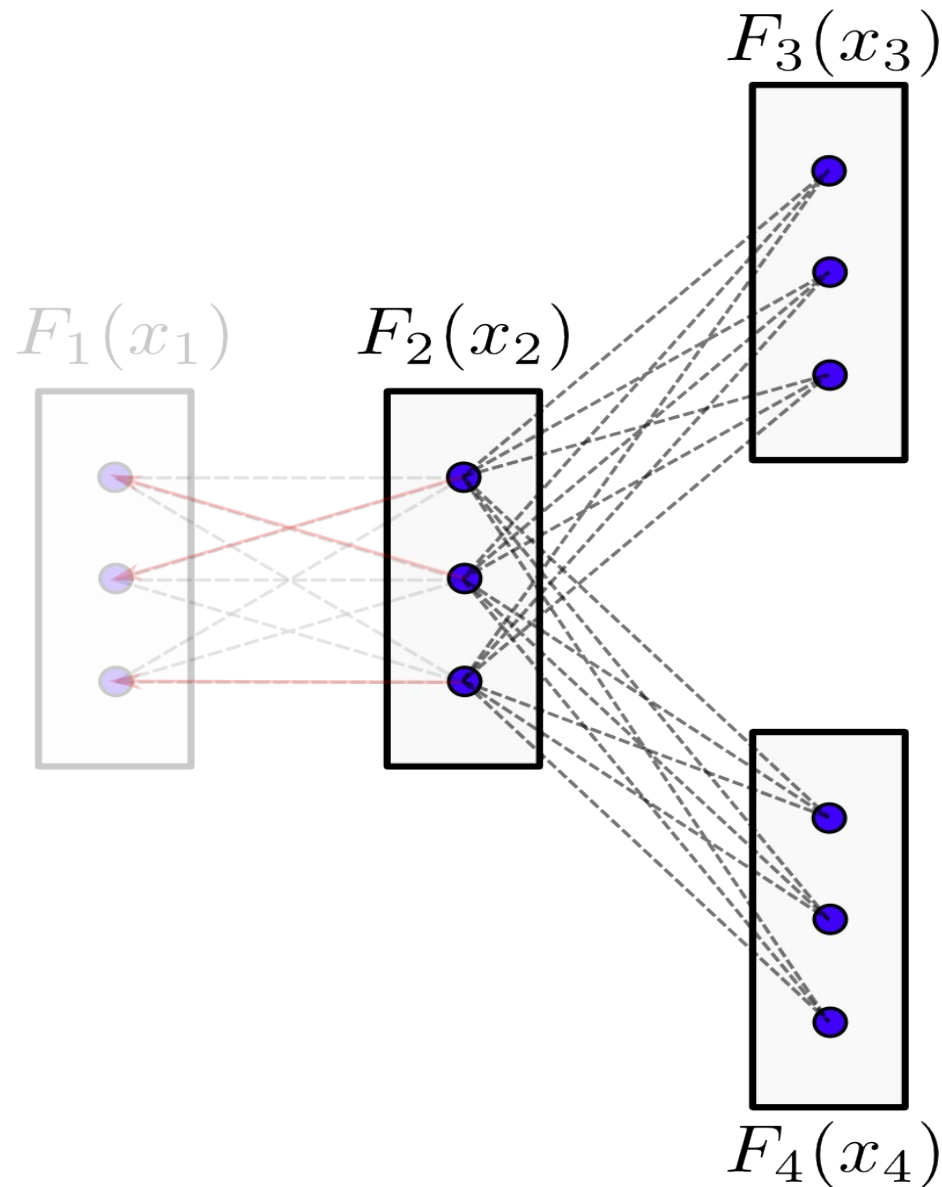
Find a leaf node:



Tree-Structured Graphical Models

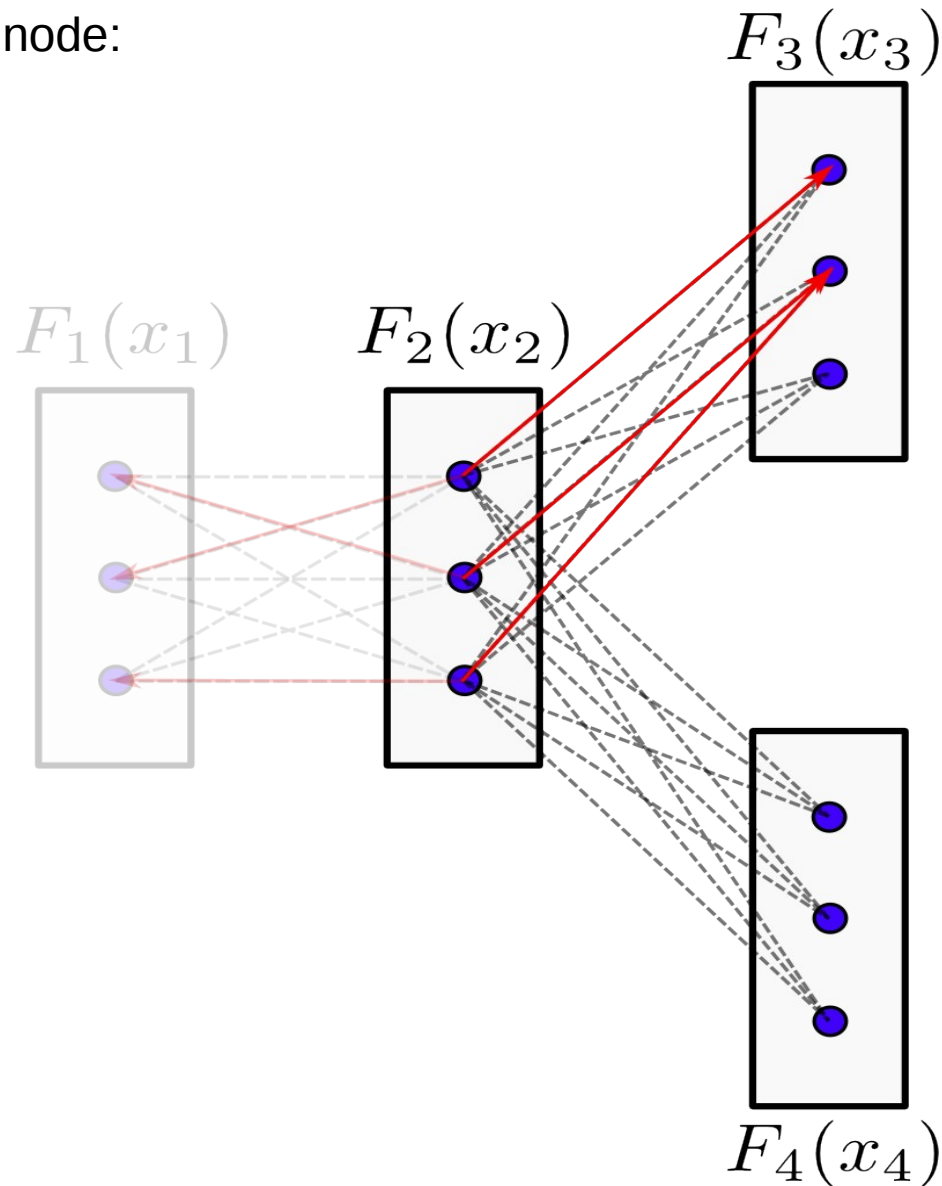


Tree-Structured Graphical Models



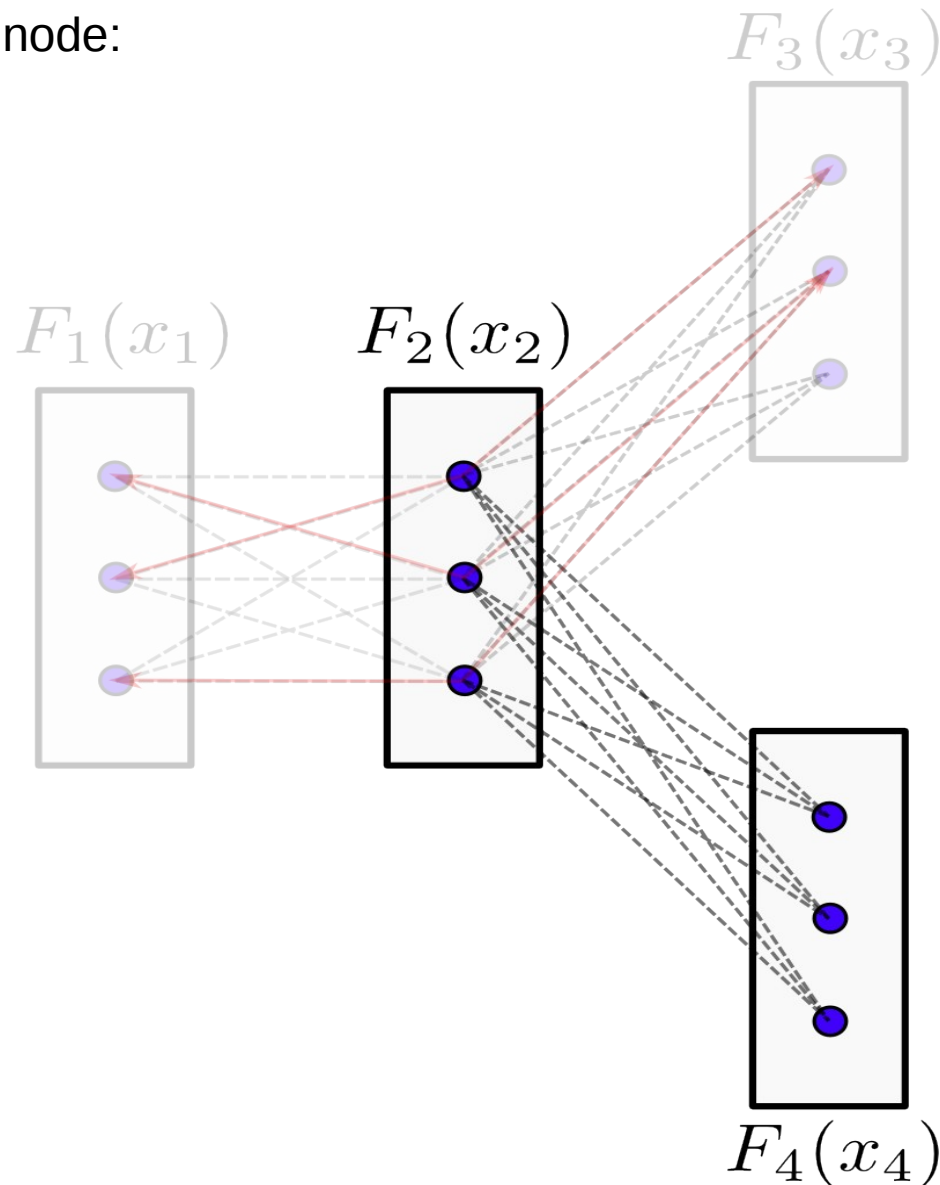
Tree-Structured Graphical Models

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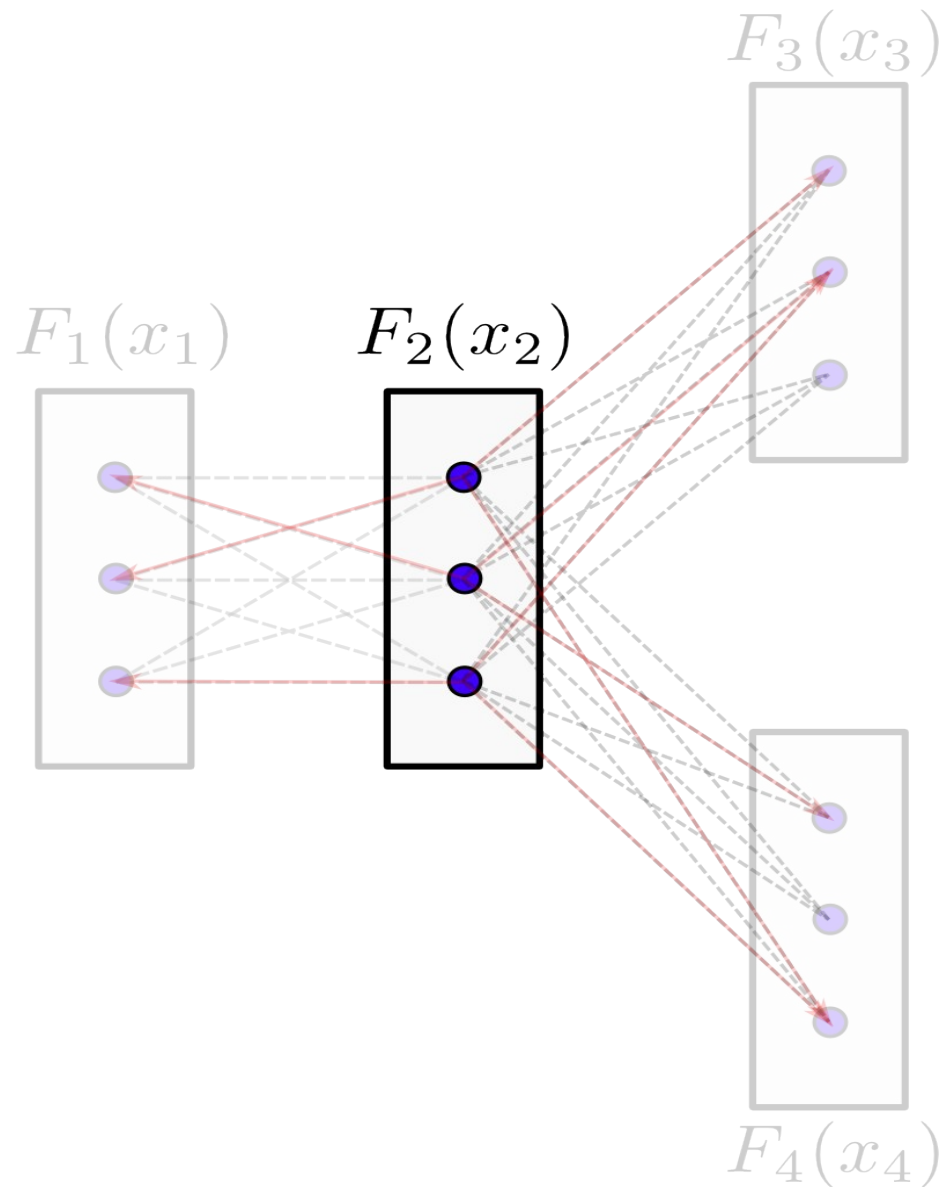


Tree-Structured Graphical Models

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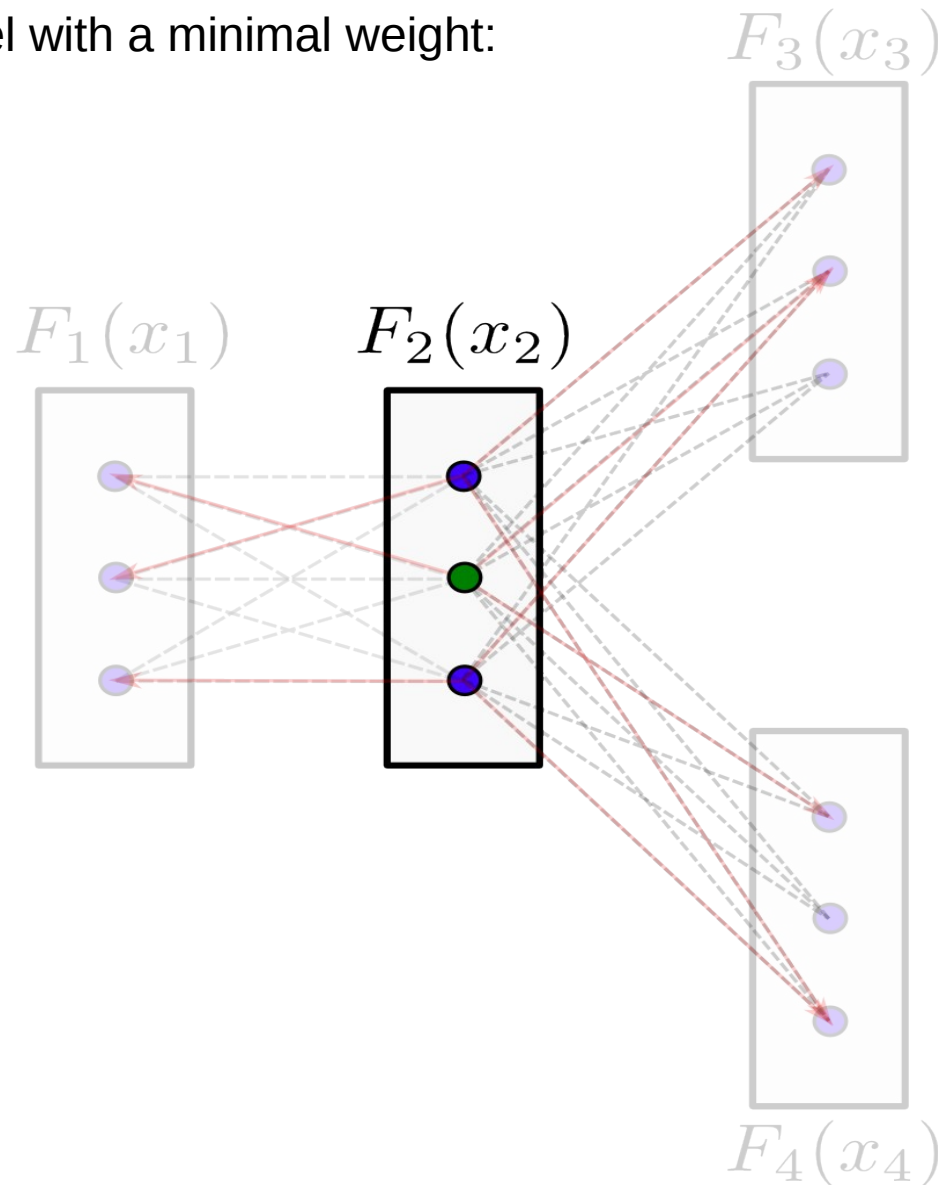


Tree-Structured Graphical Models



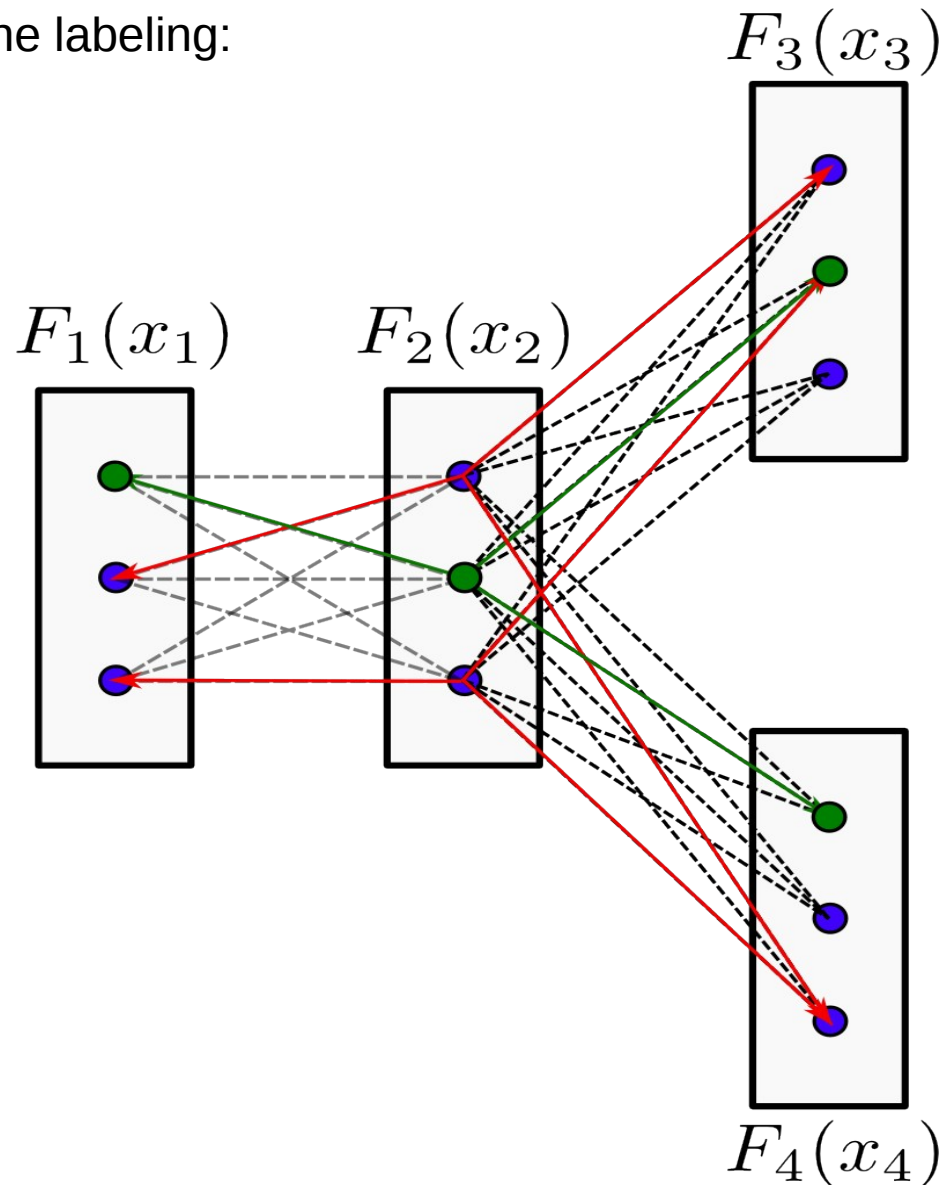
Tree-Structured Graphical Models

Select a label with a minimal weight:



Tree-Structured Graphical Models

Track back the labeling:



Acyclic Graphical Models

Why acyclic problems are so much easier?

$O(|X_0|^2|V|)$ - linear complexity instead of exponential

We will show soon that for acyclic graphical models:

$$\mathcal{L} = \mathcal{M}$$

DP finds an optimal solution of a **dual** local polytope problem

Duality theory at the next lecture!